Assignment for Week 9

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Package Loading

```r
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(haven)
```

Assignment

1) 5.1 (p 284)

(a) Since M_n has variance $\frac{1}{n}$, to make the standard deviation of M_n at most 1cm, or 0.01m, we require

$$\sqrt{\frac{1}{n}} \leq 0.01,$$

and so $n \geq 10000$. n should at least be 10000 so that the standard deviation of M_n is at most 1 centimeter.

(b) By Chebyshev's inequality,

$$P(|M_n - h| \geq 0.05) \leq \frac{\frac{1}{n}}{0.05^2}.$$

Since we want $P(|M_n - h| \geq 0.05) \leq 0.01$, we need $\frac{\frac{1}{n}}{0.05^2} \leq 0.01$. Hence $n \geq 40000$. n should at least be 40000 so that Chebyshev's inequality guarantees that the estimate is within 5 centimeters from h, with probability at least 0.99.

(c) Since the the height is bounded by $a = 1.4$, $b = 2.0$, we have

$$\sigma^2 \leq \frac{(b-a)^2}{4} = 0.6.$$

Thus to have

$$P(|M_n - h| \geq 0.05) \leq \frac{\sigma^2}{0.05^2} \leq 0.01,$$

we need $n \geq 3600$. n should at least be 3600 so that the estimate is within 5 centimeters from h, with probability at least 0.99.

2) 5.8 (p 290) The outcome of 100 rounds is distributed binomial with mean $\mu = np = 50$ and variance $\sigma^2 = np(1-p) = 25$, if the roulette is fair ($p = 0.5$).
The probability of obtaining more than 55 odd outcomes when the roulette is fair is hence

$$P(Z > \frac{55-50}{\sqrt{25}}) = P(Z > 1) = 1 - 0.8413 = 0.1587.$$

Hence the probability of making a wrong decision is 0.1587.

3) 9.20 (p 518) Since

$$Z = \frac{S_3 - n\mu_0}{\sigma_0\sqrt{n}} = \frac{S_3 - 60}{4\sqrt{3}},$$

we have $S_3 = 4\sqrt{3}Z + 60$. If the probabililty of false rejection is 0.05, $P(S_3 > \gamma) = 0.05$, and $P(4\sqrt{3}Z + 60 > \gamma) = 0.05$. Since $\Phi^{-1}(0.95)$=1.64, we have

$$\gamma = 4\sqrt{3} \times 1.64 + 60 = 71.40.$$

Therefore $\gamma = 71.40$ so that the probability of false rejection is 0.05.
The probability of false acceptance is

$$\begin{aligned}
P(S_3 < \gamma,\, H_1) &= P(Z < \frac{\gamma - n\mu_0}{\sigma_1\sqrt{n}}) \\
&= P(Z < \frac{71.40 - 60}{5\sqrt{3}}) \\
&= \Phi(1.32) \\
&= 0.90658.
\end{aligned}$$

Note that if $\mu_1 = 25$, then the probability of false acceptance is

$$\begin{aligned}
P(S_3 < \gamma,\, H_1) &= P(Z < \frac{\gamma - n\mu_1}{\sigma_1\sqrt{n}}) \\
&= P(Z < \frac{71.40 - 75}{5\sqrt{3}}) \\
&= \Phi(-0.42) \\
&= 0.3386.
\end{aligned}$$

4) 9.22 (a) (b) (p 519)

(a) By the De Moivre-Laplace approximation to the Binomial, we have for a random variable S_n distributed binomial with number of trials n and rate of success p

$$P(k \leq S_n \leq l) \approx \phi(\frac{l+\frac{1}{2}-np}{\sqrt{np(1-p)}}) - \phi(\frac{k+\frac{1}{2}-np}{\sqrt{np(1-p)}})$$

The probability of false rejection is $P(X \geq k_n; H_0)$, $\mu_0 = \frac{n}{2}$ and $\sigma_0 = \sqrt{n}\sqrt{0.5(1-0.5)} = 0.5\sqrt{n}$

$$P(S_n \geq k_n) = 1 - P(S_n \leq k_n) = 1 - \phi(\frac{k_n - \frac{1}{2} - \frac{n}{2}}{\sqrt{n(\frac{1}{2})^2}})$$
$$= 1 - \phi(\frac{k_n - \frac{1}{2} - \frac{n}{2}}{\frac{1}{2}\sqrt{n}})$$

To make $P(S_n \geq k_n) \leq 0.05$, we have

$$\begin{aligned}
&P(S_n \geq k_n) \leq 0.05 \\
&= 1 - \phi(\frac{k_n - \frac{1}{2} - \frac{n}{2}}{\frac{1}{2}\sqrt{n}}) \leq 0.05 \\
&\Rightarrow \frac{k_n - \frac{1}{2} - \frac{n}{2}}{\frac{1}{2}\sqrt{n}} = \phi^{-1}(0.95) \\
&\Rightarrow \frac{k_n - \frac{1}{2} - \frac{n}{2}}{\frac{1}{2}\sqrt{n}} = 1.645 \\
&\Rightarrow k_n = 0.8225\sqrt{n} + \frac{n+1}{2}
\end{aligned}$$

b) The probability of false acceptance is $P(X \leq k_n; H_1)$, $\mu_1 = \frac{3n}{5}$ and $\sigma_0 = \sqrt{\frac{3}{5}(1-\frac{3}{5})} = \frac{\sqrt{6}\sqrt{n}}{5}$

$$P(S_n \leq k_n) \leq 0.05$$
$$P\left(x \geq k_n : H_1\right) \approx 1 - \Phi\left(\frac{k_n - \frac{1}{2} - \frac{3}{5}n}{\sqrt{\frac{3}{5}\cdot\frac{2}{5}\cdot n}}\right) \geq 0.95$$
$$\Phi\left(\frac{k_n - \frac{1}{2} - \frac{3}{5}n}{\sqrt{\frac{3}{5}\cdot\frac{2}{5}\cdot n}}\right) \leq 0.05$$

If we plug in $k_n = 8.275\sqrt{n} + \frac{1}{2}n + \frac{1}{2}$, we get

$$\frac{0.8275\sqrt{n} - \frac{1}{10}n}{\frac{\sqrt{6n}}{5}} \leq -1.645$$

$n \geq 265.12$, meaning that $\widehat{n} = 266$ is the smallest number that satisfies the conditions.

5) 9.26 (p 519)

(a) We have the estimator $\hat{\mu}$

$$\hat{\mu} = \frac{8.47 + 10.91 + 10.87 + 9.46 + 10.40}{5} = 10.022,$$

and the estimator $\hat{\sigma}^2$

$$\hat{\sigma^2} = \frac{\sum_n^1 (x_i - \hat{u})^2}{n-1} = 1.09$$

(b) We test the statistic $\frac{\hat{\mu}-\mu}{\sigma/\sqrt{n}}$ against $t_4(0.05) = 2.132$. Since

$$\frac{|\mu - \hat{\mu}|}{\frac{\sigma}{\sqrt{n}}}$$
$$= \frac{|9 - 10.02|}{\frac{1.044}{\sqrt{5}}} \approx 2.1847 > 2.132,$$

we reject the hypothesis $\mu = 9$.

6) 9.27 (p 519) Answer: With sample size $n = 10$, the difference of sample means $\overline{X} - \overline{Y}$ is distributed normally with mean $\mu = \mu_x - \mu_y$ and variance $\sigma^2 = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{n}$.
We test the statistic $\frac{|\overline{x}-\overline{y}|}{\sqrt{\sigma}}$ against $\Phi^{-1}(0.975) = 1.96$, where $1.975 = 1 - \frac{1-0.95}{2}$. Since

$$\frac{|\overline{x} - \overline{y}|}{\sqrt{\sigma}} = \frac{181 - 177}{\sqrt{\frac{32}{10} + \frac{29}{10}}} = 1.62 < 1.92,$$

the data support t he hypothesis at the 95% significance level.

7) 9.30 (p 520)

(a) If H_0 holds, then X is distributed normally with mean $\mu = 0$ and variance $v = \sigma^2 = 0.5$. We test the statistic $\frac{X-\mu}{\sigma/\sqrt{n}}$ against $\Phi^{-1}(0.95) = 1.64$. Since

$$\frac{X - \mu}{\sigma/\sqrt{n}} = \frac{0.96 - 0}{\sqrt{0.5}} = 1.36 < 1.64,$$

we don't reject H_0 at the 5% level of significance.

(b) Now we have $n = 5$, and

$$\overline{X} = \frac{0.96 + (-0.34) + 0.85 + 0.51 + (-0.24)}{5} = 0.348,$$

we test the statistic $\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$ against $\Phi^{-1}(0.95) = 1.64$. Since

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{0.348 - 0}{\sqrt{0.5}\sqrt{5}} = 1.1 < 1.64,$$

we don't reject H_0 at the 5% level of significance.

(c) Since v is unknown, we compute the estimator $\hat{\sigma}$ of sample variance:

$$\hat{\sigma}^2 = \frac{(0.96 - 0.348)^2 + (-0.34 - 0.348)^2 + (0.85 - 0.348)^2 + (0.51 - 0.348)^2 + (-0.24 - 0.348)^2}{4} = 0.368,$$

and so $\hat{\sigma} = \sqrt{0.3680} = 0.61$.
We test the statistic $\frac{\overline{X}-\mu}{\hat{\sigma}/\sqrt{n}}$ against $t_4^{-1}(0.95) = 2.132$. Since

$$\frac{\overline{X} - \mu}{\hat{\sigma}/\sqrt{n}} = \frac{0.348 - 0}{0.61\sqrt{5}} = 1.28 < 2.132,$$

we don't reject H_0 at the 5% level of significance using the t-distribution.

8) Suppose $\overline{X}$ is the mean of 100 observations from a popualtion with mean μ and variance σ^2. Find limits between which $\overline{X} - \mu$ will lie with a probability of at least .85 using the Central Limit Theorem.
Answer: we want to find k such that

$$P(\frac{|\overline{X} - \mu|}{\sigma\sqrt{n}} < k) = 0.85,$$

where $n = 100$. For the standard normal distribution Z, if

$$P(|Z| < k) = 0.85,$$

then $\Phi(k) = 1 - \frac{1-0.85}{2} = 0.925$, and $k = \Phi^{-1}(0.925) = 1.44$. Therefore,

$$P(\frac{|\overline{X} - \mu|}{\sigma\sqrt{n}} < 1.44) = 0.85,$$

$$P(-\frac{1.44\sigma}{10} < |\overline{X} - \mu| < \frac{1.44\sigma}{10}) = 0.85$$

and the desired limits between which $\overline{X} - \mu$ will lie with a probability of at least .85 is $[-\frac{1.44\sigma}{10}, \frac{1.44\sigma}{10}]$.

9)

(a) Verify that $f(x,y)$ os a PDF.
Proof: $f(x,y)$ is a valid PDF because

$$0 \leq f(x_i, y_j) \leq 1 \quad \forall x_i \in \{1,2,3\} \text{ and } y_j \in \{1,2,3\},$$

and

$$\sum_{i=1}^{3}\sum_{j=1}^{3} f(x_i, y_j) = 0.1 + 0 + 0.1 + 0 + 0.2 + 0.2 + 0.1 + 0.2 + 0.1 = 1.$$

(b) Find the marginal probability functions of X and Y.

$$f(x=1) = \sum_{j=1}^{3} f(x=1, y_j) = 0.1 + 0 + 0.1 = 0.2,$$
$$f(x=2) = \sum_{j=1}^{3} f(x=2, y_j) = 0 + 0.2 + 0.2 = 0.4,$$
$$f(x=3) = \sum_{j=1}^{3} f(x=3, y_j) = 0.1 + 0.2 + 0.1 = 0.4.$$

Therefore, the marginal probability function of X is

$$f_X(x) = \begin{cases} 0.2 & \text{if } x = 1, \\ 0.4 & \text{if } x = 2, \\ 0.4 & \text{if } x = 3. \end{cases}$$

Similarly,

$$f(y=1) = \sum_{i=1}^{3} f(x_i, y=1) = 0.1 + 0 + 0.1 = 0.2,$$
$$f(y=2) = \sum_{i=1}^{3} f(x_i, y=1) = 0 + 0.2 + 0.2 = 0.4,$$
$$f(y=3) = \sum_{i=1}^{3} f(x_i, y=3) = 0.1 + 0.2 + 0.1 = 0.4.$$

Therefore, the marginal probability function of Y is

$$f_Y(y) = \begin{cases} 0.2 & \text{if } y = 1, \\ 0.4 & \text{if } y = 2, \\ 0.4 & \text{if } y = 3. \end{cases}$$

(c) Compute $E[X]$ and $V(X)$.

$$\begin{aligned}E[X] &= \sum_{i=1}^{3} x f_X(x)\\ &= 1\times 0.2 + 2\times 0.4 + 3\times 0.4\\ &= 2.2\end{aligned}$$

$$\begin{aligned}E[X^2] &= \sum_{i=1}^{3} x^2 f_X(x)\\ &= 1\times 0.2 + 2^2\times 0.4 + 3^2\times 0.4\\ &= 5.4\end{aligned}$$

Thus we have

$$\begin{aligned}V(X) &= E[X^2] - E[X]^2\\ &= 5.4 - 2.2^2\\ &= 0.56.\end{aligned}$$

(d) Compute the covariance between X and Y.
We have

$$\begin{aligned}E[XY] &= \sum_{i=1}^{3} xyf(x,y)\\ &= 1\times 0.1 + 2\times 0 + 3\times 0.1 + 2\times 0 +\\ &\quad 4\times 0.2 + 6\times 0.2 + 3\times 0.1 + 6\times 0.2 + 9\times 0.1\\ &= 4.8,\end{aligned}$$

and $E(Y) = E(X) = 2.2$ because of symmetry, thus the covariance $Cov(X,Y)$ between X and Y is

$$Cov(X,Y) = E[XY] - E[X]E[Y] = 4.8 - 2.2\times 2.2 = -0.04.$$

(e) Find P(X<Y).

$$\begin{aligned}P(X<Y) &= P(X=1,Y=2) + P(X=1,Y=3) + P(X=2,Y=3)\\ &= 0 + 0.1 + 0.2\\ &= 0.3.\end{aligned}$$

(f) Are X and Y independent? Justify your answer.
If X and Y independent, $Cor(X,Y) = 0$. Since

$$Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{V(X)V(Y)}} = \frac{-0.04}{0.56} = -0.071 \neq= 0,$$

X and Y are not independent.

10) Write your own functional equivalents to the listed values of each of the following functions. Each should take on a vector and output one of the listed values.

a) `t.test(X,Y, var.equal=TRUE)` - statistic, parameter, p.value, conf.int.
b) `var.test(X,Y)` - statistic, parameter, p.value, conf.int.

Answer: The functional equivalent I use for a) is the t-test, and for b) the F-test. I also compare the result with the built-in functions in R, namely `t.test` and `var.test`.

```r
set.seed(120)

a <- rnorm(200)
b <- rnorm(200, mean=0.5)
#By coding like above, the variances of the two samples are equal to 0.

#a) t-test
dummytstat <- function(a,b){
abar <- mean(a)
bbar <- mean(b)
sda <- sd(a)
sdb <- sd(b)
vara<-var(a)
varb<-var(b)
na<-length(a)
nb<-length(b)
t.st <-(abar-bbar)/sqrt(vara/na+varb/nb)
return(t.st)
}

dummytstat(a,b)
```

```
## [1] -4.747723
```

```r
#Comparison with t-test function
t.test(a, b, var.equal=TRUE, conf.level = 0.95)
```

```
##
##  Two Sample t-test
##
## data:  a and b
## t = -4.7477, df = 398, p-value = 2.875e-06
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.7053514 -0.2922596
## sample estimates:
##  mean of x  mean of y
## 0.03852436 0.53732986
```

```r
#b) F-test
set.seed(120)
a <- rnorm(200)
b <- rnorm(200, mean=0.5)

dummyfstat <- function(a,b){
vara <- var(a)
varb <- var(b)
```

```
dofa <-length(a)-1
dofb <-length(b)-1

fstat <- vara/varb

pval <- pf(fstat, dofa, dofb)*2
#2 is for two-tail
ans <- paste ("F-Stat", fstat, "p-val(2tail)", pval)

return(ans)
}

dummyfstat(a,b)
```

```
## [1] "F-Stat 0.844566770585627 p-val(2tail) 0.234308088702764"
```

```
#Comparison with var.test function
var.test(a, b, conf.level = 0.95)
```

```
##
##  F test to compare two variances
##
## data:  a and b
## F = 0.84457, num df = 199, denom df = 199, p-value = 0.2343
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.6391568 1.1159907
## sample estimates:
## ratio of variances
##          0.8445668
```